

Sum of numbers from 0 to x (including x) using a for loop

```
def sumNumbersForLoop(x):  
    total = 0      # Create a variable to hold the sum  
    for loopVariable in range(x+1): # Loop [0, 1, ..., x]  
        total += loopVariable # Add loopVariable to sum  
    return total   # Return the output of the function
```

```
output = sumNumbersForLoop(4)  
print(output)
```

10

```
# The above call does the following:  
# output = sumNumbersForLoop(4)  
# sumNumbersForLoop(4)  
#   x = 4  
#   total = 0  
#   loopVariable = 0  
#   total = total + loopVariable = 0 + 0 = 0  
#   loopVariable = 1  
#   total = total + loopVariable = 0 + 1 = 1  
#   loopVariable = 2  
#   total = total + loopVariable = 1 + 2 = 3  
#   loopVariable = 3  
#   total = total + loopVariable = 3 + 3 = 6  
#   loopVariable = 4  
#   total = total + loopVariable = 6 + 4 = 10  
#   return 10  
# output = 10  
# print(output)
```

Recursion basics

In recursion, a function **calls** itself. Recursion is used when a problem can be easily divided into **easier** problems.

A recursive function has **2 components**:

1. **Base case:** Simplest possible input and prevents **infinite recursion**.
2. **Recursion step:** Call the same function itself with a **smaller/easier** input to the function and act on the output of the smaller function call.

Sum of numbers from 0 to x using recursion

```
def sumNumbersRecurse(x):  
    if x == 0:                # Base case, simplest possible number  
        return 0  
  
    sum_zero_to_xminus1 = sumNumbersRecurse(x-1) # Recursion step,  
    with simpler input  
    total = x + sum_zero_to_xminus1  
  
    return total
```

Walk-through of code

The above call does the following (draw out on the board):

```
output = sumNumbersRecurse(4)
```

First, `sumNumbersRecurse` calls itself

```
sumNumbersRecurse(4) = sumNumbersRecurse(3) + 4  
sumNumbersRecurse(3) = sumNumbersRecurse(2) + 3  
sumNumbersRecurse(2) = sumNumbersRecurse(1) + 2  
sumNumbersRecurse(1) = sumNumbersRecurse(0) + 1  
sumNumbersRecurse(0) = 0
```

When `sumNumbersRecurse(x)` receives the output from `sumNumbersRecurse(x-1)`, it **adds** x and then returns the sum.

```
sumNumbersRecurse(1) = 0 + 1 = 1  
sumNumbersRecurse(2) = 1 + 2 = 3  
sumNumbersRecurse(3) = 3 + 3 = 6  
sumNumbersRecurse(4) = 6 + 4 = 10  
output = 10
```

DEMO TIME

Common mistakes

Forgetting the base case

```
# Let's use the factorial problem we solved together in the morning  
and see common issues.  
  
# The formula for factorial is:  
# 0! = 1  
# x! = x * (x-1) * (x-2) * ... * 1
```

```
def brokenFactorial(x):
    print("x = " + str(x) + ", call brokenFactorial(" + str(x-1) +
    ")")
    factorial_x_minus_1 = brokenFactorial(x-1)
    result = x * factorial_x_minus_1
    print("I'm about to return a result: " + str(result))
    return result
```

```
output = brokenFactorial(5)
print('output = ' + str(output))
```

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RecursionError

Traceback (most recent call
last)

Cell In[2], line 1

```
----> 1 output = brokenFactorial(5)
      2 print('output = ' + str(output))
```

Cell In[1], line 10, in brokenFactorial(x)

```
7 def brokenFactorial(x):
9     print("x = " + str(x) + ", call brokenFactorial(" + str(x-
```

```

1) + ")")
---> 10     factorial_x_minus_1 = brokenFactorial(x-1)
      11     result = x * factorial_x_minus_1
      12     print("I'm about to return a result: " + str(result))

```

```

Cell In[1], line 10, in brokenFactorial(x)
      7 def brokenFactorial(x):
      9     print("x = " + str(x) + ", call brokenFactorial(" + str(x-
1) + ")")
---> 10     factorial_x_minus_1 = brokenFactorial(x-1)
      11     result = x * factorial_x_minus_1
      12     print("I'm about to return a result: " + str(result))

```

[... skipping similar frames: brokenFactorial at line 10 (2968 times)]

```

Cell In[1], line 10, in brokenFactorial(x)
      7 def brokenFactorial(x):
      9     print("x = " + str(x) + ", call brokenFactorial(" + str(x-
1) + ")")
---> 10     factorial_x_minus_1 = brokenFactorial(x-1)
      11     result = x * factorial_x_minus_1
      12     print("I'm about to return a result: " + str(result))

```

```

Cell In[1], line 9, in brokenFactorial(x)
      7 def brokenFactorial(x):
----> 9     print("x = " + str(x) + ", call brokenFactorial(" + str(x-
1) + ")")
      10     factorial_x_minus_1 = brokenFactorial(x-1)
      11     result = x * factorial_x_minus_1

```

```

File
/opt/homebrew/lib/python3.11/site-packages/ipykernel/iostream.py:560,
in OutputStream.write(self, string)
    558     raise ValueError(msg)
    559 else:
--> 560     is_child = not self._is_master_process()
    561     # only touch the buffer in the IO thread to avoid races
    562     with self._buffer_lock:

```

```

File
/opt/homebrew/lib/python3.11/site-packages/ipykernel/iostream.py:445,
in OutputStream._is_master_process(self)
    444 def _is_master_process(self):
--> 445     return os.getpid() == self._master_pid

```

RecursionError: maximum recursion depth exceeded while calling a Python object

Forgetting to use partial answers

```
def anotherBrokenFactorial(x):  
    if x == 0:  
        print("x = " + str(x) + ", returning 1")  
        return 1  
  
    print("x = " + str(x) + ", call anotherBrokenFactorial(" + str(x-  
1) + ")")  
  
    factorial_x_minus_1 = anotherBrokenFactorial(x-1)  
  
    result = factorial_x_minus_1  
    print("x = " + str(x) + ", I'm about to return a result: " +  
str(result))  
    return result
```

```
output = anotherBrokenFactorial(5)  
print('output = ' + str(output))
```

```
x = 5, call anotherBrokenFactorial(4)  
x = 4, call anotherBrokenFactorial(3)  
x = 3, call anotherBrokenFactorial(2)  
x = 2, call anotherBrokenFactorial(1)  
x = 1, call anotherBrokenFactorial(0)  
x = 0, returning 1  
x = 1, I'm about to return a result: 1  
x = 2, I'm about to return a result: 1  
x = 3, I'm about to return a result: 1  
x = 4, I'm about to return a result: 1  
x = 5, I'm about to return a result: 1  
output = 1
```

Your Turn :)

What does this second function do?

```
def mystery2(a, b):  
    if b == 0:  
        return 0  
    return a + mystery2(a, b - 1)
```

what is the output of the line of code below?

```
mystery2(5, 4)
```

20